Categorical Data Analysis HW1

Name: 莊明儒 Student ID: 111024513

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1 Problem 1: Analysis of mtcars data

1.1 An initial data analysis that explores the numerical and graphical characteristics of the data

Data Description

The mtcars data was extracted from the 1974 Motor Trend US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles. The data has 32 observations on 11 variables. The variables are:

- `mpg`: Miles/(US) gallon
- `cyl`: Number of cylinders
- `disp`: Displacement (cubic inches)
- `hp`: Gross horsepower
- `drat`: Rear axle ratio

- `wt`: Weight (1000 lbs)
- `qsec`: 1/4 mile time
- `vs`: Engine (0=V-shaped, 1=straight)
- `am`: Transmission (0=automatic, 1=manual)
- `gear`: Number of forward gears
- `carb`: Number of carburetors

where `vs` and `am` are nominal and binary while other variables are all continuous. Table 1 shows the several basic statistics, including minimum, quantiles, maximum, mean and median, for each variable.

```
dat = read.table("mtcars.txt", fill = TRUE)
```

```
stat = round(sapply(dat, summary),2)
kable(
  as.data.frame(stat),
  format = "latex", booktabs = TRUE,
  align = "c", caption = "Summary for variables",
  escape = FALSE, linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 1: Summary for variables

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Min.	10.40	4.00	71.10	52.00	2.76	1.51	14.50	0.00	0.00	3.00	1.00
1st Qu.	15.43	4.00	120.83	96.50	3.08	2.58	16.89	0.00	0.00	3.00	2.00
Median	19.20	6.00	196.30	123.00	3.70	3.33	17.71	0.00	0.00	4.00	2.00
Mean	20.09	6.19	230.72	146.69	3.60	3.22	17.85	0.44	0.41	3.69	2.81
3rd Qu.	22.80	8.00	326.00	180.00	3.92	3.61	18.90	1.00	1.00	4.00	4.00
Max.	33.90	8.00	472.00	335.00	4.93	5.42	22.90	1.00	1.00	5.00	8.00

From Table 1, we can see several interesting facts:

- The distributions of `disp`, `hp`, and `carb` seem to be right-skewed since their means seem obviously higher than their medians in consideration of their scales.
- By looking at the quantiles of each variable, we can see that the scales seem very different among the variables.
- The proportion of automobiles with straight engine is 0.44%, and that of automobiles with manual transmission is 0.41%.
- No missing value or obviously extreme value seems to present.

Data Feature

To explore the basic relationships among the variable, Figure 1 and 2 are plotted. Accordingly, several interesting findings are listed as follows.

- The distributions of `cyl`, `disp`, `hp` and `qsec` seem quite different between the automobiles with straight engine and those with shaped engine.

- The distributions of `cyl`, `disp`, `hp`, `drat`, `wt` and `gear` seem quite different between the automobiles with straight engine and those with shaped engine.
- `mpg` and `cyl` seem to negatively associate with each other. However, they seem conditionally uncorrelated on `vs`.It is obvious that there is high collinearity between `cyl` and `vs`, and thus we are not supposed to use `cyl` and `vs` simultaneously when building model for prediction of `mpg`.
- `mpg` seems negatively correlated with `disp`, `hp`, `wt` and `carb` and slightly negatively associated with `carb`.
- `mpg` seems positively correlated with `drat` and `qsec` and slighty positively associated with `gear`.
- The `mpg` of the automobiles with straight engine seems higher than those with shaped engine.
- The `mpg` of the automobiles with manual transmission seems higher than those with automatic transmission.

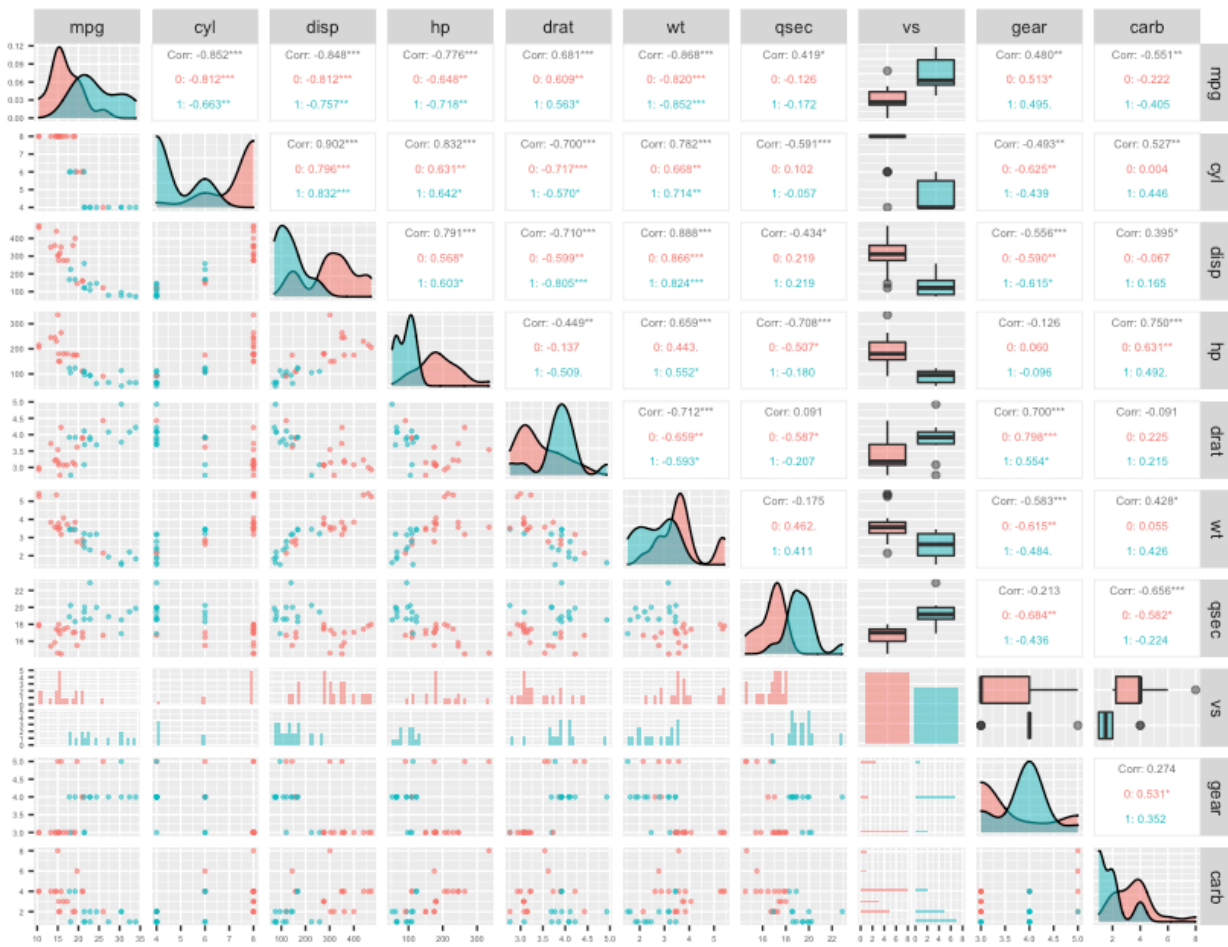

Figure 1: Relatinships among variables under two levels of `vs`

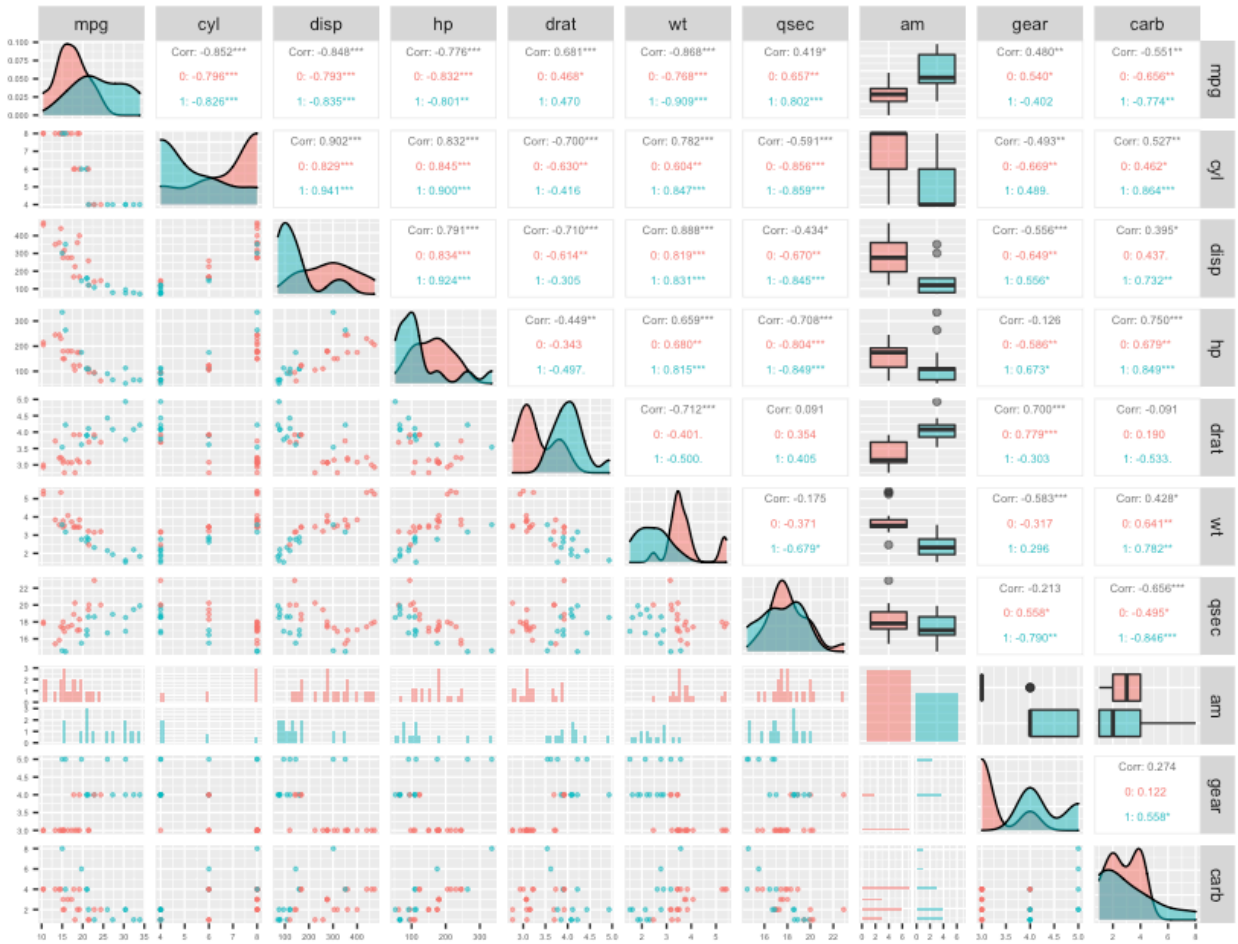

Figure 2: Relatinships among variables under two levels of `am`

1.2 Model fitting

According to the findings from the exploration above, we decide to first postulate a model as follows.

$$\texttt{mpg} = \beta_0 + \beta_1\texttt{disp} + \beta_2\texttt{hp} + \beta_3\texttt{drat} + \beta_4\texttt{wt} + \beta_5\texttt{qsec} + \beta_6\texttt{vs} + \beta_7\texttt{am} + \beta_8\texttt{gear} + \beta_9\texttt{carb} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Table 2 shows the fitting result of the postulated model. It is found that none of the variable is significant at 0.05 significance level. This may be due to the fact that there are so many parameters to estimate in the current model.

```
fit = lm(mpg~.-cyl, dat)
summ.table(fit)
```

Therefore, we remove the variable with the least significance in terms of the p-value step by step. Notice that only the one with the largest p-value would be removed in each step. Such process of variable selection is illustrated in Appendix A in details.

Table 2: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	10.9601	13.5303	0.8100	0.4266
disp	0.0128	0.0168	0.7626	0.4538
hp	-0.0219	0.0209	-1.0477	0.3062
drat	0.8352	1.5363	0.5437	0.5921
wt	-3.6925	1.8395	-2.0073	0.0572
qsec	0.8424	0.6868	1.2267	0.2329
vs1	0.3897	1.9480	0.2001	0.8433
am1	2.5774	1.9403	1.3283	0.1977
gear	0.7116	1.3656	0.5210	0.6075
carb	-0.2196	0.7886	-0.2785	0.7833

```
fit = update(fit, .~.-vs); summ.table(fit)
fit = update(fit, .~.-carb); summ.table(fit)
fit = update(fit, .~.-gear); summ.table(fit)
fit = update(fit, .~.-drat); summ.table(fit)
fit = update(fit, .~.-disp); summ.table(fit)
fit1 = update(fit, .~.-hp); summ.table(fit)
```

```
summ.table(fit1)
```

Table 3: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	9.6178	6.9596	1.3819	0.1779
wt	-3.9165	0.7112	-5.5069	0.0000
qsec	1.2259	0.2887	4.2467	0.0002
am1	2.9358	1.4109	2.0808	0.0467

It turns out that only `wt`, `qsec` and `am` are significant. The model is thus revised as follows.

$$\texttt{mpg} = \beta_0' + \beta_1'\texttt{wt} + \beta_2'\texttt{qsec} + \beta_3'\texttt{am} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I). \tag{1}$$

This model will be referred as Model (1) in the later pages.

1.3 An exploration of transformations to improve the fit of the model

Consider to apply Box-Cox transformation on `mpg`, the response variable, i.e., to decide a suitable value of λ such that the model

$$\frac{\texttt{mpg}^\lambda - 1}{\lambda} = \gamma_0 + \gamma_1\texttt{disp} + \gamma_2\texttt{hp} + \gamma_3\texttt{drat} + \gamma_4\texttt{wt} + \gamma_5\texttt{qsec} + \gamma_6\texttt{vs} + \gamma_7\texttt{am} + \gamma_8\texttt{gear} + \gamma_9\texttt{carb} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

is most plausible in terms of the maximized log-likelihood. Figure 3 shows the log-likelihood function of λ, where we can see that the log-likelihood function is nearly maximized at $\lambda = 0$. Since

$$\lim_{\lambda \to 0} \frac{\texttt{mpg}^{\lambda} - 1}{\lambda} = \log(\texttt{mpg}),$$

it is suggested to postulate the model as follows.

$$\log(\texttt{mpg}) = \gamma_0 + \gamma_1 \texttt{disp} + \gamma_2 \texttt{hp} + \gamma_3 \texttt{drat} + \gamma_4 \texttt{wt} + \gamma_5 \texttt{qsec} + \gamma_6 \texttt{vs} + \gamma_7 \texttt{am} + \gamma_8 \texttt{gear} + \gamma_9 \texttt{carb} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I).$$

```
boxcox(fit1)
```

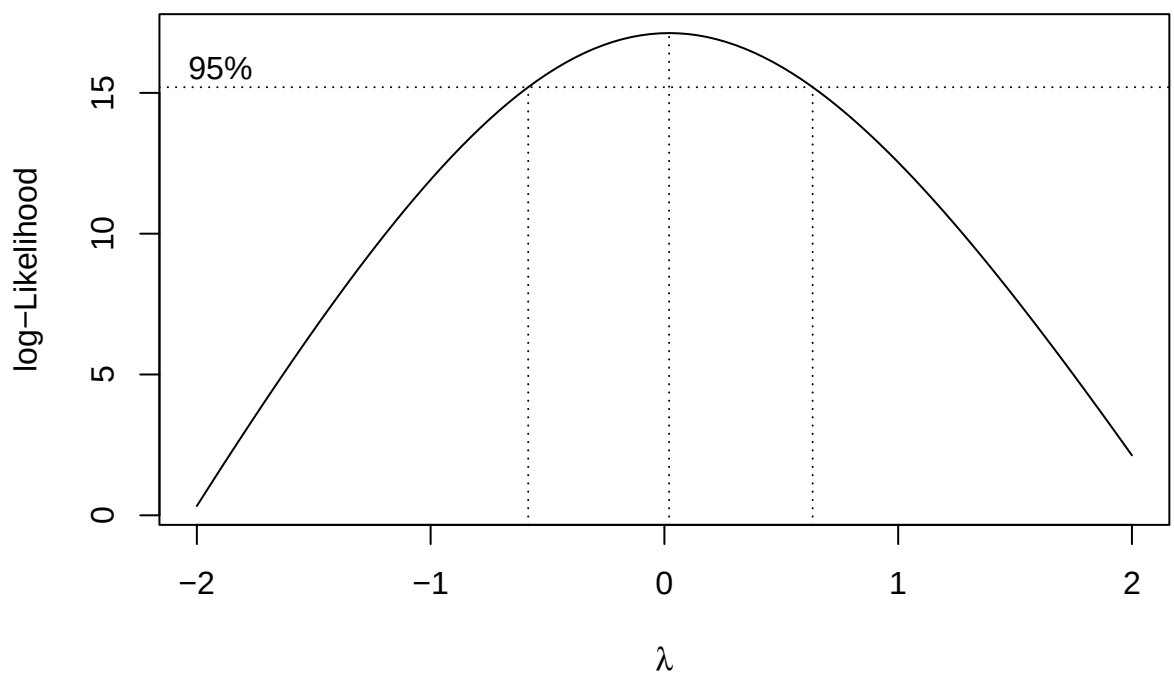

Figure 3: Log-likelihood function of λ

Table 4 shows the fitting result of the postulated model. It is found that only `wt` is significant at 0.05 significance level. This may be due to the fact that there are so many parameters to estimate in the current model.

```
fit = lm(log(mpg)~.-cyl, dat)
summ.table(fit)
```

Therefore, we remove the variable with the least significance in terms of the p-value step by step. Notice that only the one with the largest p-value would be removed in each step. Such process of variable selection is illustrated in Appendix B in details.

Table 4: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	2.8679	0.6141	4.6701	0.0001
disp	0.0001	0.0008	0.1109	0.9127
hp	-0.0009	0.0009	-0.9136	0.3708
drat	0.0189	0.0697	0.2710	0.7889
wt	-0.1739	0.0835	-2.0827	0.0491
qsec	0.0293	0.0312	0.9399	0.3575
vs1	-0.0078	0.0884	-0.0884	0.9303
am1	0.0434	0.0881	0.4934	0.6267
gear	0.0554	0.0620	0.8937	0.3811
carb	-0.0187	0.0358	-0.5234	0.6059

```
fit = update(fit, .~.-vs); summ.table(fit)
fit = update(fit, .~.-disp); summ.table(fit)
fit = update(fit, .~.-drat); summ.table(fit)
fit = update(fit, .~.-am); summ.table(fit)
fit = update(fit, .~.-carb); summ.table(fit)
fit = update(fit, .~.-gear); summ.table(fit)
fit2 = update(fit, .~.-qsec); summ.table(fit)
```

```
summ.table(fit2)
```

Table 5: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	3.8291	0.0687	55.7522	0e+00
hp	-0.0015	0.0004	-3.9791	4e-04
wt	-0.2005	0.0272	-7.3778	0e+00

It turns out that only `wt`, `qsec` and `am` are significant. The model is thus revised as follows.

$$\texttt{mpg} = \gamma_0' + \gamma_1'\texttt{hp} + \gamma_2'\texttt{wt} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I). \tag{2}$$

This model will be referred as Model (2) in the later pages. Table 6 shows the comparison between Model (1) and Model (2). Since the R^2 of Model (2) is greater than that of Model (1) and σ of Model (2) is smaller than that of Model (1), we conclude that Model (2) leads to a better fit of the data. In other words, the log transformation applied on `mpg` seems helpful in terms of fitting.

```
compare = round(matrix(c(summary(fit1)$r.squared, summary(fit1)$sigma,
         summary(fit2)$r.squared, summary(fit2)$sigma), ncol = 2), 4)
colnames(compare) = c("Model (1)", "Model (2)")
rownames(compare) = c("$R^2$", "$\\sigma$")
kable(compare, align = "c", escape = FALSE, booktabs = TRUE,
      caption = "Model Comparison")
```

Table 6: Model Comparison

	Model (1)	Model (2)
R^2	0.8497	0.8691
σ	2.4588	0.1114

1.4 Diagnostics to check the assumptions of my model

1.4.1 Model diagnostics for Model (1)

Diagnostic for leverage point

```
n = nrow(dat)
p = 4
```

```
lev = lm.influence(fit1)$hat
plot(lev, ylab = "Leverages", main = "index plot of leverages")
abline(h = 2*p/n, col = "red")
```

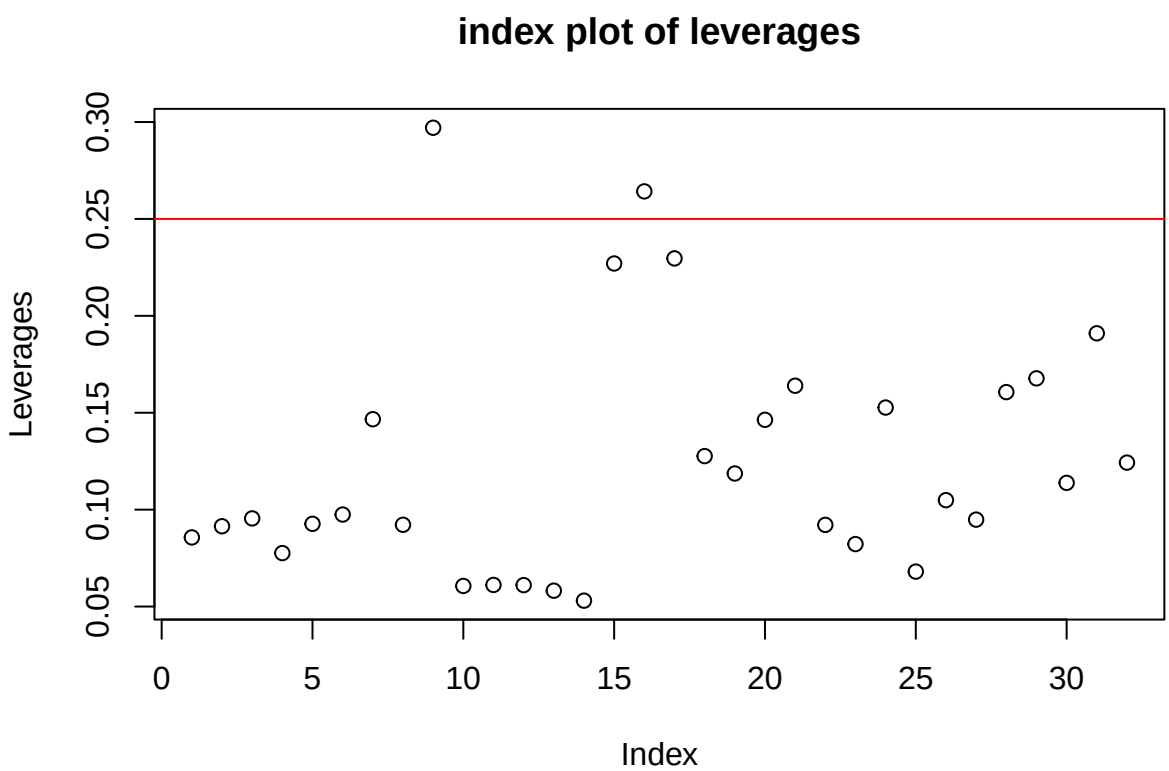

Figure 4: Plot for leverage of each observation

Figure 4 shows the leverage of each observation. A horizontal line at $2p/n$, which is decided by the rule of thumb, is marked on the plot to help identify those with too large leverages.

```r
lev[lev > 2*p/n]
```

```
##            Merc.230 Lincoln.Continental
##           0.2970422           0.2642151
```

It is found that Merc 230 and Lincoln Continental seem to have their leverages too large, and thus we identify them as the leverage points.

Diagnostic for outlier

```r
jack = rstudent(fit1)
plot(jack, ylab="Jackkniife residuals", main="Jackkniife residuals")
```

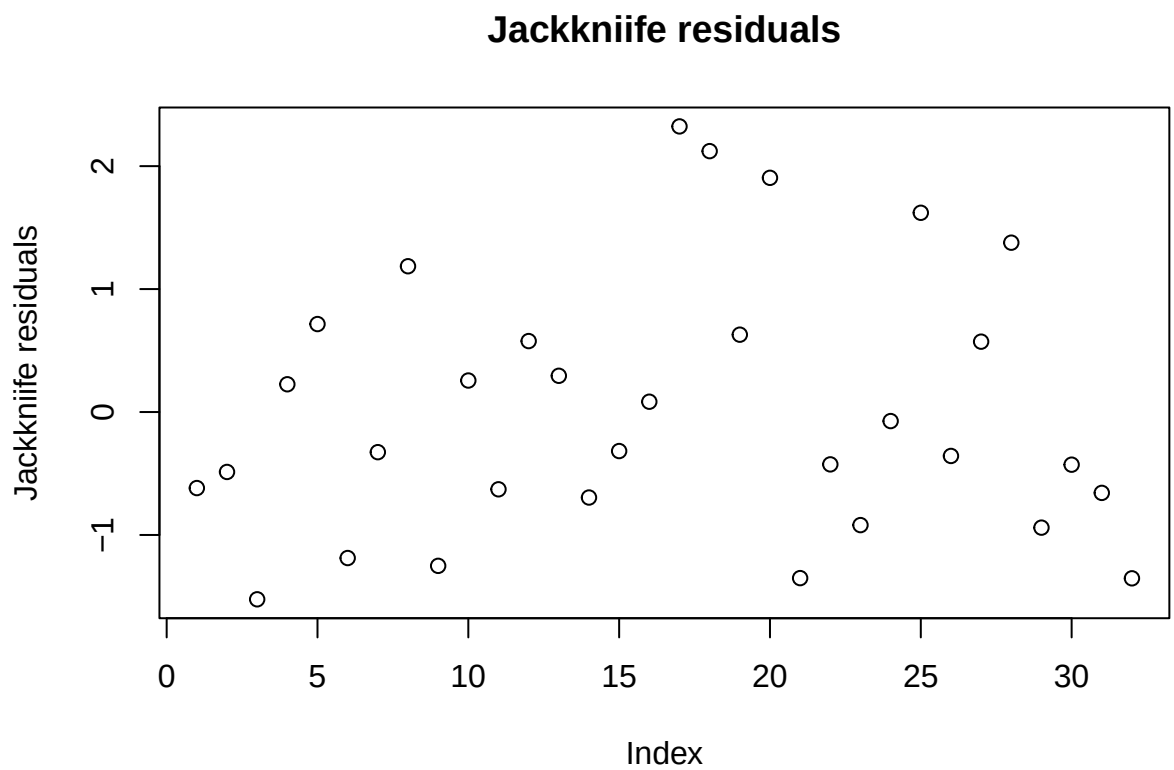

Figure 5: Plot for Jackknife residuals

From Figure 5, it seems that none of the observations looks like an outlier. However, it is still better to go through some numerical procedure for more objective checking.

It can be shown that the Jackkniife residuals follow t distribution with degrees of freedom $n-p-1$ under null if the model is correct and $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Hence hypothesis testing described as follows can be conducted to examine if their is any outlier present in the data.

$$\begin{cases} H_0 : \text{No outlier is present.} \\ H_1 : \text{There is at least one outlier.} \end{cases}$$

```
critical = qt(.05/n, n-p-1, lower.tail = FALSE)
```

Note that this proposed test is involved with multiple testing. Bonferroni's correction is one of the measures to deal with such problem. Since the sample size $n = 32$ and the number of parameters $p = 4$, we have the critical value for this test $t_{27}^{0.05/32} = 3.2450836$ given significance level 0.05. We reject H_0 whenever any of the Jackkniife residual exceeds 3.2450836.

```
jack[abs(jack) > critical]
```

```
## named numeric(0)
```

It turns out that none of the residuals exceeds the critical value, and thus we conclude that we do not have strong evidence to say there is some outlier present in the data. This result is consistent with our conjecture when looking at the Jackkniife residual plot.

Diagnostic for influential point

Cook's distance is used to examine whether there is any influential point.

```
cook = cooks.distance(fit1)
plot(cook, ylab = "Cooks distance")
```

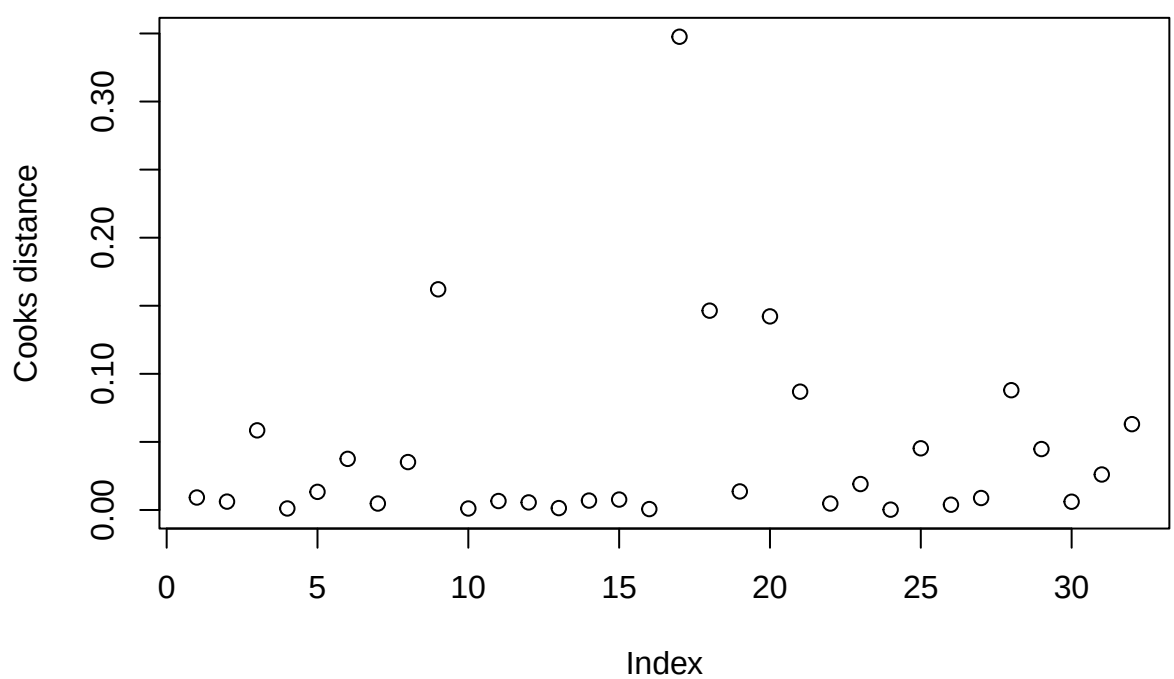

Figure 6: Cook's distance of each observation

Figure 6 shows that some of the observations might have their Cook's distances too large. Such observations seem to be influential points.

```
names(sort(cook, decreasing = TRUE)[1:3])
```

```
## [1] "Chrysler.Imperial" "Merc.230"          "Fiat.128"
```

It is found that Chrysler Merc 230 and Fiat 128 have the largest influence on the fitting result. We decide to look more deeply to see if there is still any other observation which would have strong influence on the estimation for some specific coefficients.

```
fit1inf = lm.influence(fit1)
```

```
inf.ls = list()
par(mfrow = c(2, 2))
variables = c("wt", "qsec", "am")
for(i in 1:length(variables)){
  plot(fit1inf$coef[,i+1], ylab = paste("change in ", variables[i], " coef", sep = ""))
  abline(h = 0)
  inf.ls[[i]] = names(sort(fit1inf$coef[,i+1], decreasing = TRUE)[1:3])
}
plot(fit1inf$sigma, ylab =  expression(paste("change in ",hat(sigma)), sep = ""))
abline(h = 0)
```

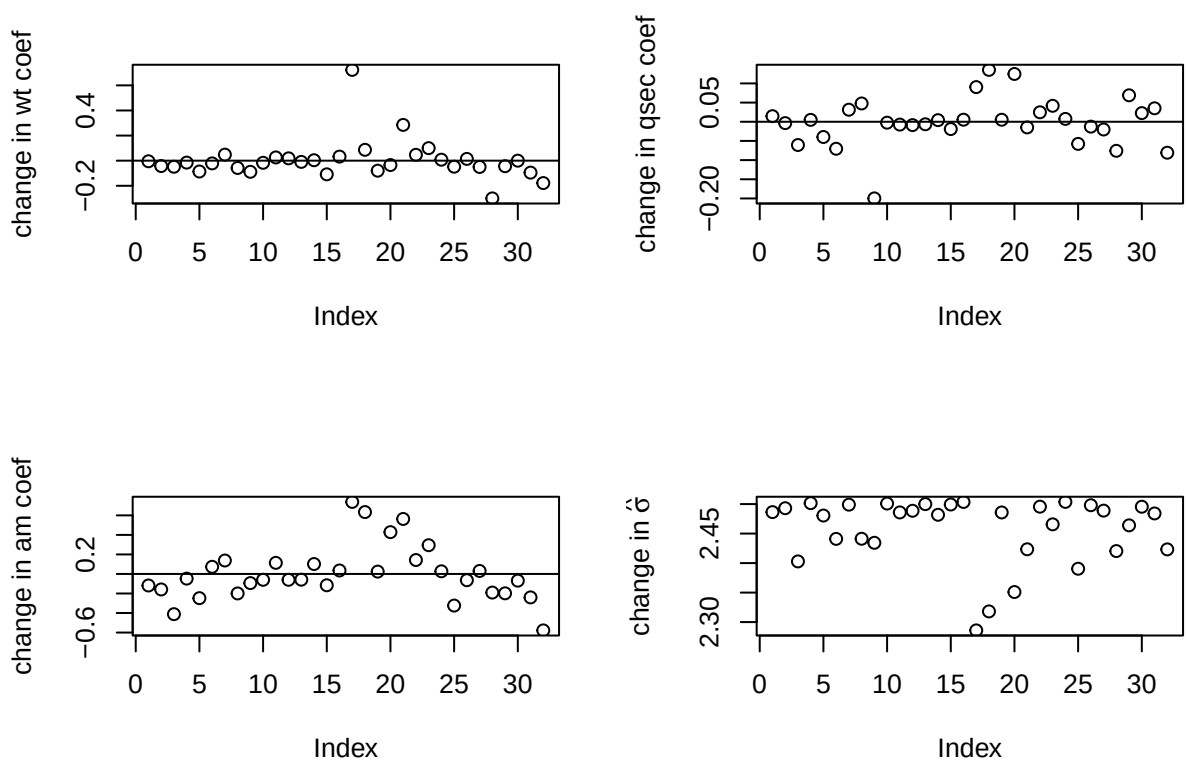

Figure 7: Changes on coefficient estimates due to each observation

Table 7: Summary for the influential points

	Chrysler.Imperial	Toyota.Corona	AMC.Javelin	Fiat.128
wt				
qsec	*			*
am	*	*		*
$\hat{\sigma}$				

Table 8: Summary for the influential points (Cont.)

	Toyota.Corolla	Camaro.Z28	Lincoln.Continental	Hornet.4.Drive
wt				
qsec	*			
am				
$\hat{\sigma}$		*	*	*

```
inf.ls[[length(variables) + 1]] = names(sort(fit1inf$sigma, decreasing = TRUE)[1:3])
```

Figure 7 shows how each coefficient and $\hat{\sigma}$ would change when leaving one of the observations out to fit the model. The larger the change means that the observation that is left out would have higher influence on the parameter estimate. We see that there indeed seems to exist some influential points for the parameter estimates. Table 8 summarizes which observation might have relatively high influence on the parameter estimates, where "*" denotes that the corresponding observation is among the first three most influential points for the corresponding parameter estimate.

```
summ = matrix("", ncol = length(unique(unlist(inf.ls))),
              nrow = length(variables) + 1)
colnames(summ) = unique(unlist(inf.ls))
rownames(summ) = c(variables, "$\\hat{\\sigma}$")
for(i in 1:length(variables) + 1){
  fill = rep("", ncol(summ))
  fill[unique(unlist(inf.ls)) %in% inf.ls[[i]]] = "*"
  summ[i,] = fill
}
kable(summ[,1:4], caption = "Summary for the influential points",
      escape = FALSE, booktabs = TRUE, align = "c")
```

```
kable(summ[,5:8], caption = "Summary for the influential points (Cont.)",
      escape = FALSE, booktabs = TRUE, align = "c")
```

Though there seems to be some sample points having relatively large influences, we choose not to remove any of them, considering that the change on the coefficient estimates due to their influences does not seem really big compared to the coefficient estimates obtained with the full data.

Diagnostic for nonconstant variance

```
lev = lm.influence(fit1)$hat
rstud = fit1$res/(summary(fit1)$sigma*(1-lev))
par(mfrow = c(2, 2))
```

```
plot(fit1$res,ylab="raw residuals")
plot(rstud,ylab="studentized residuals")
plot(jack,ylab="Jackkniife residuals")
```

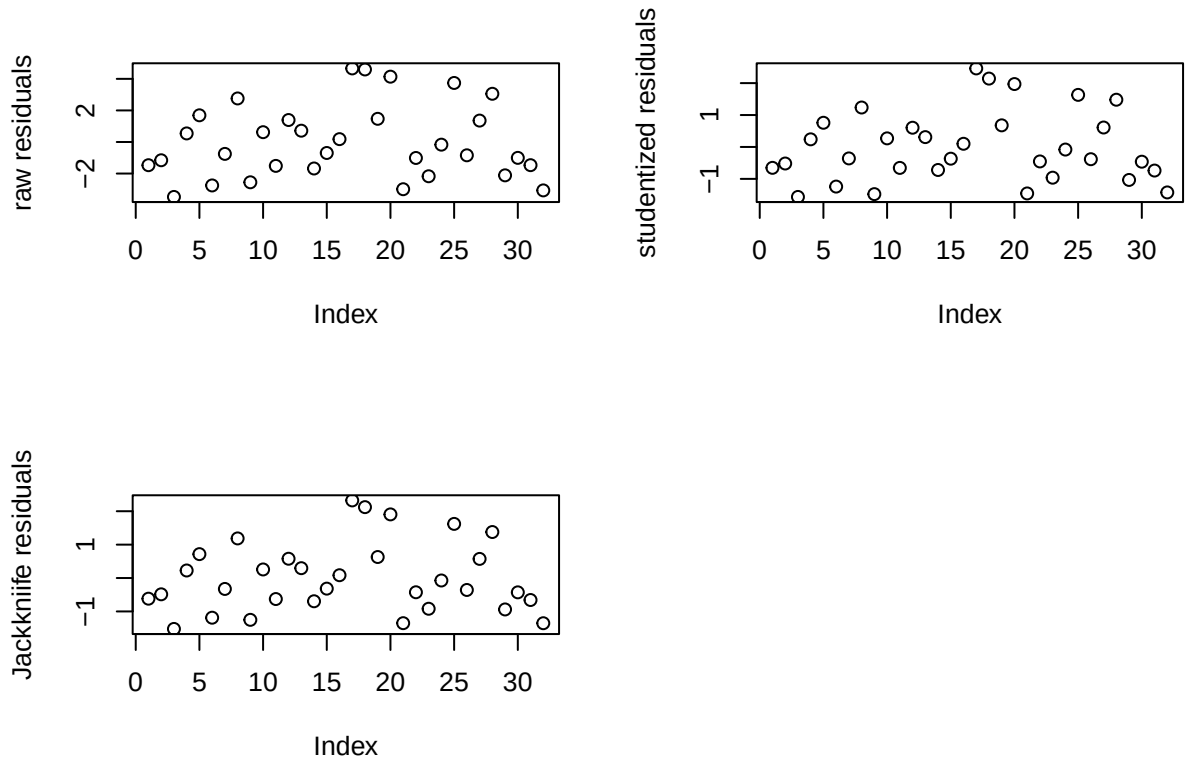

Figure 8: Residual plots using raw residuals, studentized residuals and Jackknife residuals

Figure 8 shows that the patterns of the three kinds of residual plots look similar. Therefore, we decide to use the raw residuals in the following analysis. To increase the resolution of detecting non-constant variance, the plot of absolute residuals against fitted values is shown in Figure 9.

```
plot(abs(fit1$res)~fit1$fitted.values, ylab="|raw residuals|", xlab = "fitted value")
```

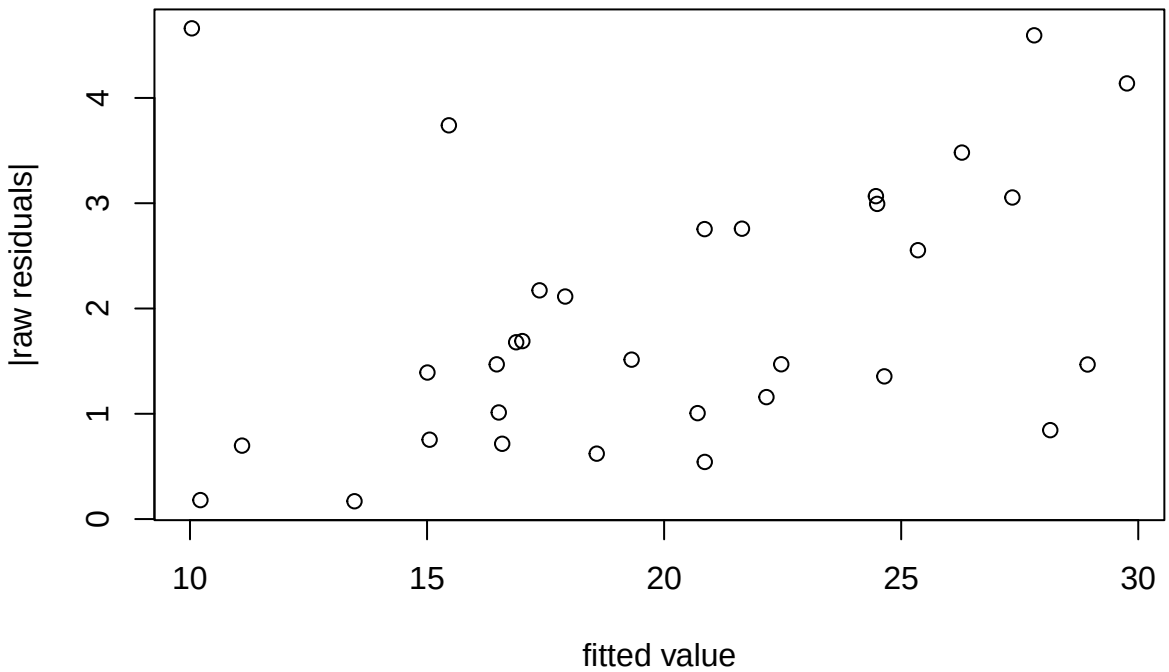

Figure 9: Absolute residuals against fitted values

It seems that we should take care about the problem of nonconstant variance. We can roughly make a quick check for it by regressing the absolute residuals on the fitted values. The fitting result for the regression just described is shown in Table 9. We can see that the coefficient of "fitted values" is significant under 0.05 significance level. Hence, we conclude that the assumption for constant variance might not be acceptable.

```
nonconst.fit = lm(abs(fit1$res)~fit1$fitted.values)
summ.table(nonconst.fit, cap = "Test for nonconstant variance by regression",
           rows = c("(Intercept)", "fitted values"))
```

Table 9: Test for nonconstant variance by regression

	Estimate	SE	t-value	p-value
(Intercept)	0.2605	0.8080	0.3224	0.7494
fitted values	0.0832	0.0388	2.1439	0.0403

Diagnostic for curvature in the mean of residuals

```
par(mfrow = c(2, 2))

d = lm(mpg~qsec+am, data = dat)$res
m = lm(wt~qsec+am, data = dat)$res
```

```
plot(m, d, xlab = "wt residuals", ylab = "mpg residual", main = "added residual plot")

d = lm(mpg~wt+am, data = dat)$res
m = lm(qsec~wt+am, data = dat)$res
plot(m, d, xlab = "qsec residuals", ylab = "mpg residual", main = "added residual plot")

d = lm(mpg~wt+qsec, data = dat)$res
m = lm(as.numeric(am)~wt+qsec, data = dat)$res
plot(m, d, xlab = "am residuals", ylab = "mpg residual", main = "added residual plot")
```

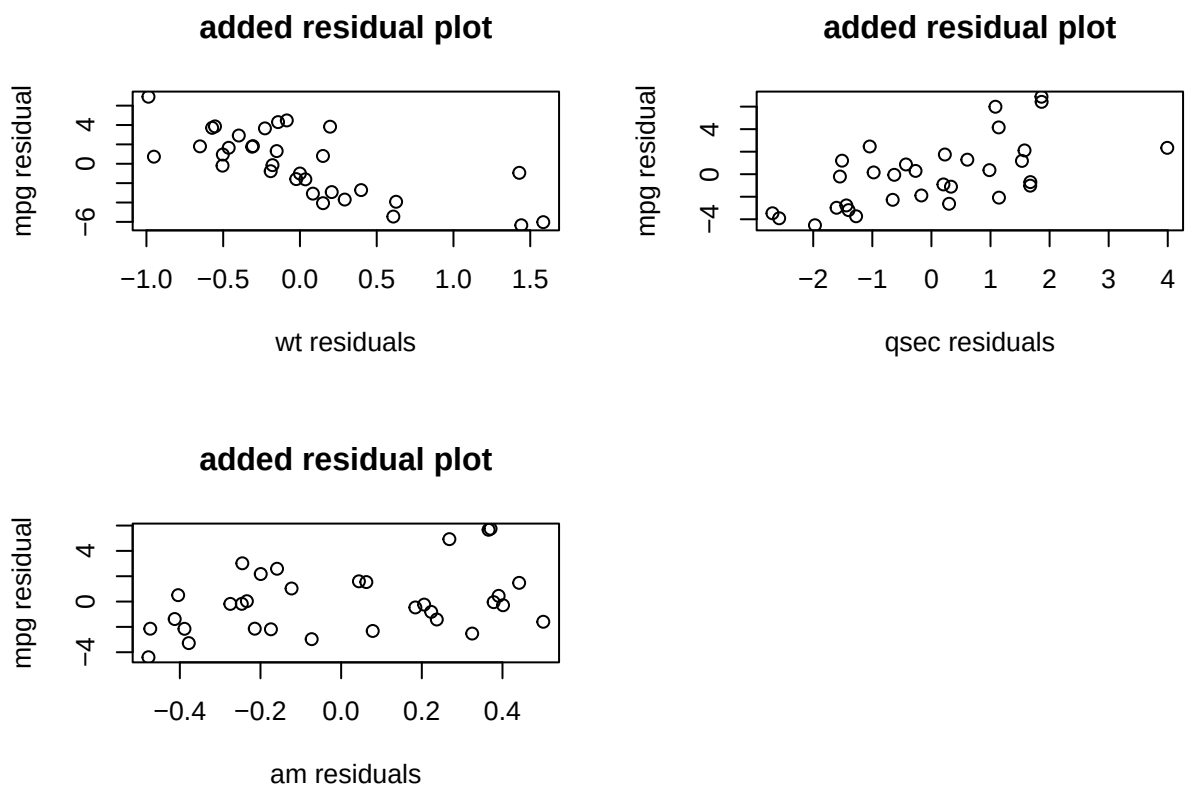

Figure 10: Added variable plots

Figure 10 shows the added residual plot for each predictor. Since each plot shows that there seems to exist only linear relationship between the two kinds of residuals, we conclude that there might not be curvature in the mean of residuals.

Diagnostic for Normality

```
qqnorm(fit1$res, ylab = "raw residuals", main = "")
qqline(fit1$res)
```

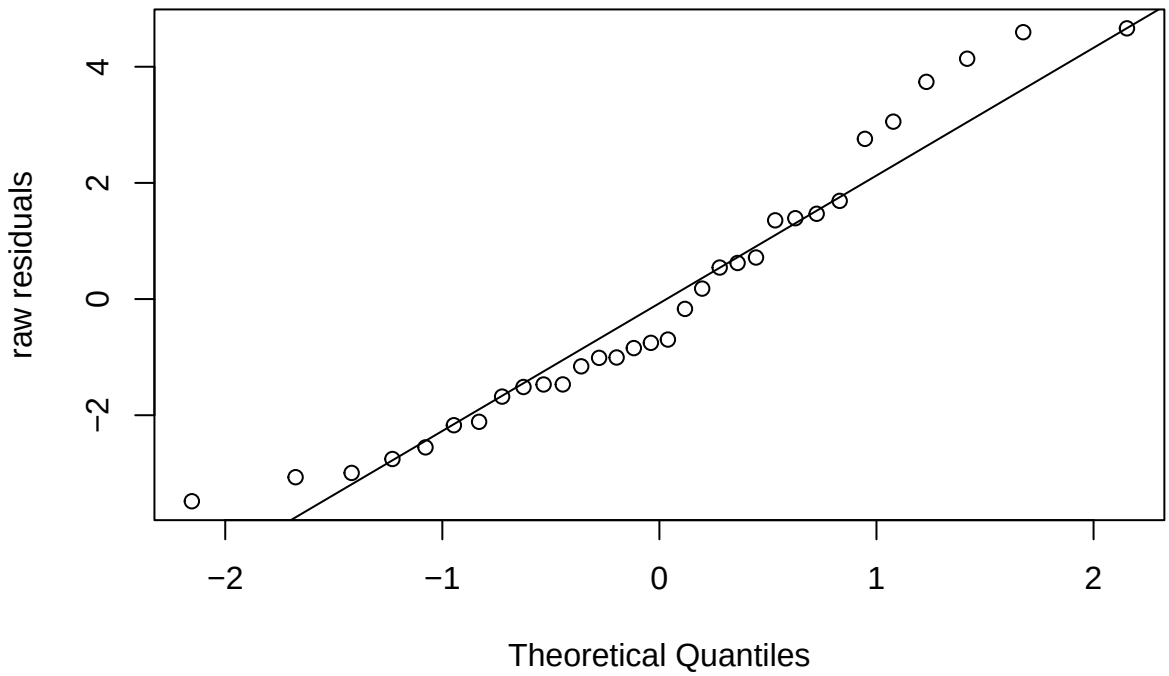

Figure 11: Normal Q-Q plot

Q-Q plot is used to check for normality. In Figure 11, since all the points lie quite close to the line joining the first and the third quartiles, we conclude that there is no strong evidence to say that the normality assumption does not hold. In other words, normality might be an acceptable assumption.

1.4.2 Model diagnostics for Model (2)

Diagnostic for leverage point

```
n = nrow(dat)
p = 3
```

```
lev = lm.influence(fit2)$hat
plot(lev, ylab = "Leverages", main = "index plot of leverages")
abline(h = 2*p/n, col = "red")
```

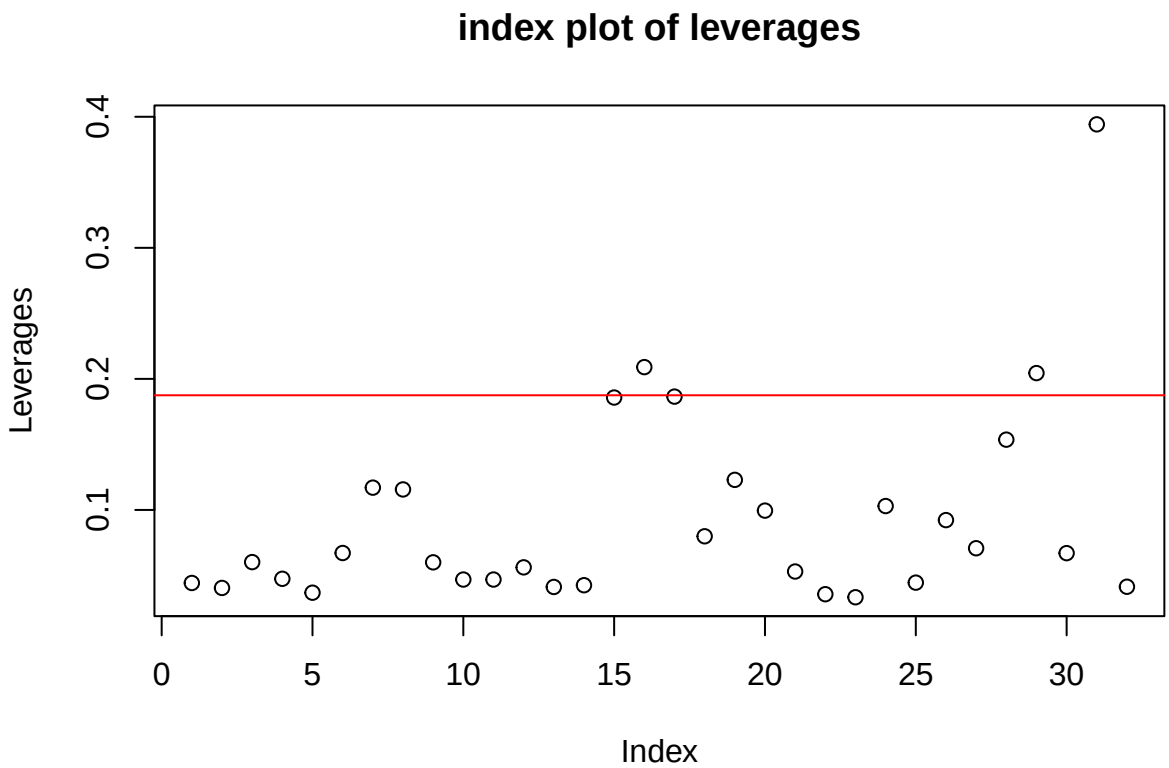

Figure 12: Plot for leverage of each observation

Figure 12 shows the leverage of each observation. A horizontal line at $2p/n$, which is decided by the rule of thumb, is marked on the plot to help identify those with too large leverages.

```
lev[lev > 2*p/n]
```

```
## Lincoln.Continental      Ford.Pantera.L       Maserati.Bora
##           0.2089784           0.2044239           0.3942082
```

It is found that Lincoln Continental, Ford Pantera L and Maserati Bora seem to have their leverages too large, and thus we identify them as the leverage points.

Diagnostic for outlier

```
jack = rstudent(fit2)
plot(jack, ylab="Jackkniife residuals", main="Jackkniife residuals")
```

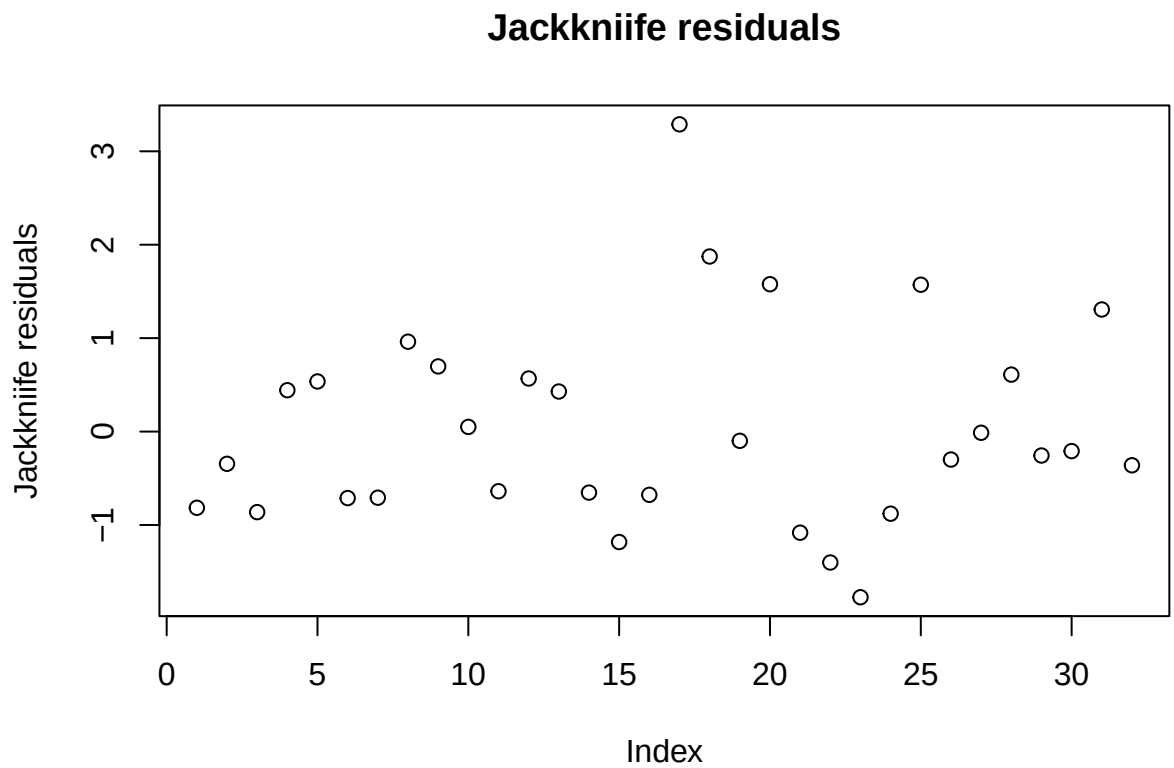

Figure 13: Plot for Jackknife residuals

From Figure 13, it seems quite difficult to say whether any outlier is present. It is better to go through some numerical procedure for an objective checking.

It can be shown that the Jackkniife residuals follow t distribution with degrees of freedom $n-p-1$ under null if the model is correct and $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Hence hypothesis testing described as follows can be conducted to examine if their is any outlier present in the data.

$$\begin{cases} H_0 : \text{No outlier is present.} \\ H_1 : \text{There is at least one outlier.} \end{cases}$$

```
critical = qt(.05/n, n-p-1, lower.tail = FALSE)
```

Note that this proposed test is involved with multiple testing. Bonferroni's correction is one of the measures to deal with such problem. Since the sample size $n = 32$ and the number of parameters $p = 3$, we have the critical value for this test $t_{28}^{0.05/32} = 3.2338659$ given significance level 0.05. We reject H_0 whenever any of the Jackkniife residual exceeds 3.2338659.

```
jack[abs(jack) > critical]
```

```
## Chrysler.Imperial
##          3.288354
```

It turns out that the residual of Chrysler Imperial exceeds the critical value, and thus we conclude that Chrysler.Imperial might be an outlier.

Diagnostic for influential point

Cook's distance is used to examine whether there is any influential point.

```
cook = cooks.distance(fit2)
plot(cook, ylab = "Cooks distance")
```

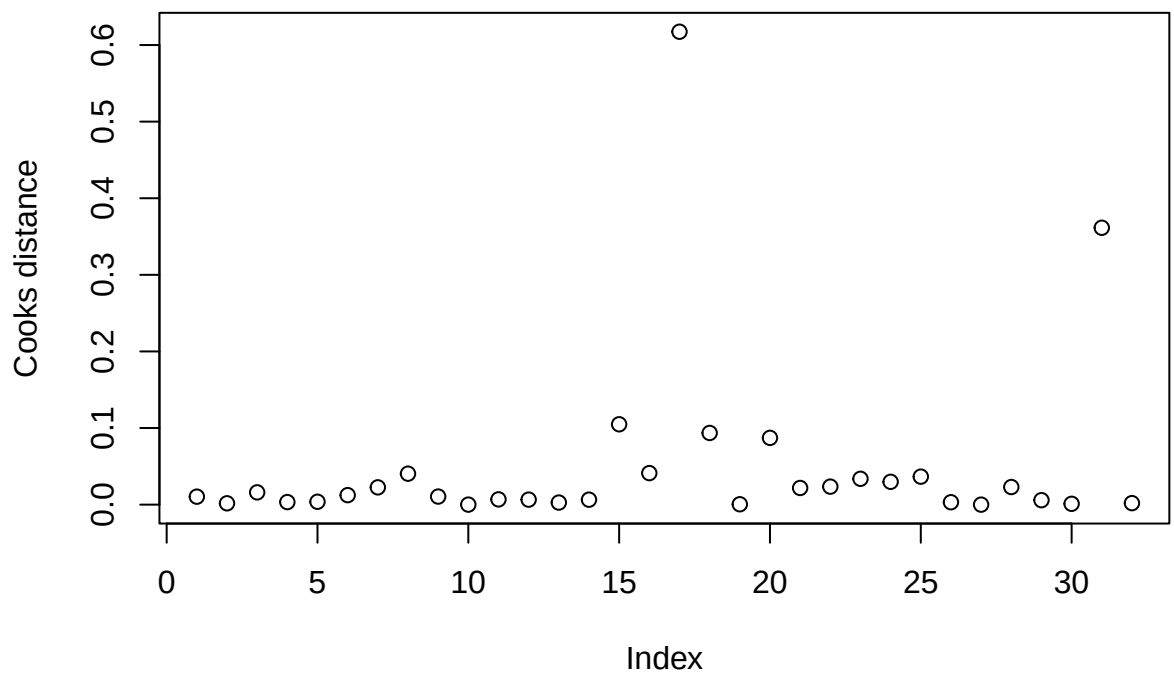

Figure 14: Cook's distance of each observation

Figure 14 shows that some of the observations might have their Cook's distances too large. Such observations seem to be influential points.

```
names(sort(cook, decreasing = TRUE)[1:3])
```

```
## [1] "Chrysler.Imperial"  "Maserati.Bora"      "Cadillac.Fleetwood"
```

It is found that Chrysler Imperial, Maserati Bora and Cadillac Fleetwood have the largest influence on the fitting result. We decide to look more deeply to see if there is still any other observation which would have strong influence on the estimation for some specific coefficients.

```r
fit2inf = lm.influence(fit2)
```

```r
inf.ls = list()
par(mfrow = c(2, 2))
variables = c("hp", "wt")
for(i in 1:length(variables)){
  plot(fit2inf$coef[,i+1], ylab = paste("change in ", variables[i], " coef", sep = ""))
  abline(h = 0)
  inf.ls[[i]] = names(sort(fit2inf$coef[,i+1], decreasing = TRUE)[1:3])
}
plot(fit2inf$sigma, ylab =  expression(paste("change in ",hat(sigma)), sep = ""))
abline(h = 0)
inf.ls[[length(variables) + 1]] = names(sort(fit2inf$sigma, decreasing = TRUE)[1:3])
```

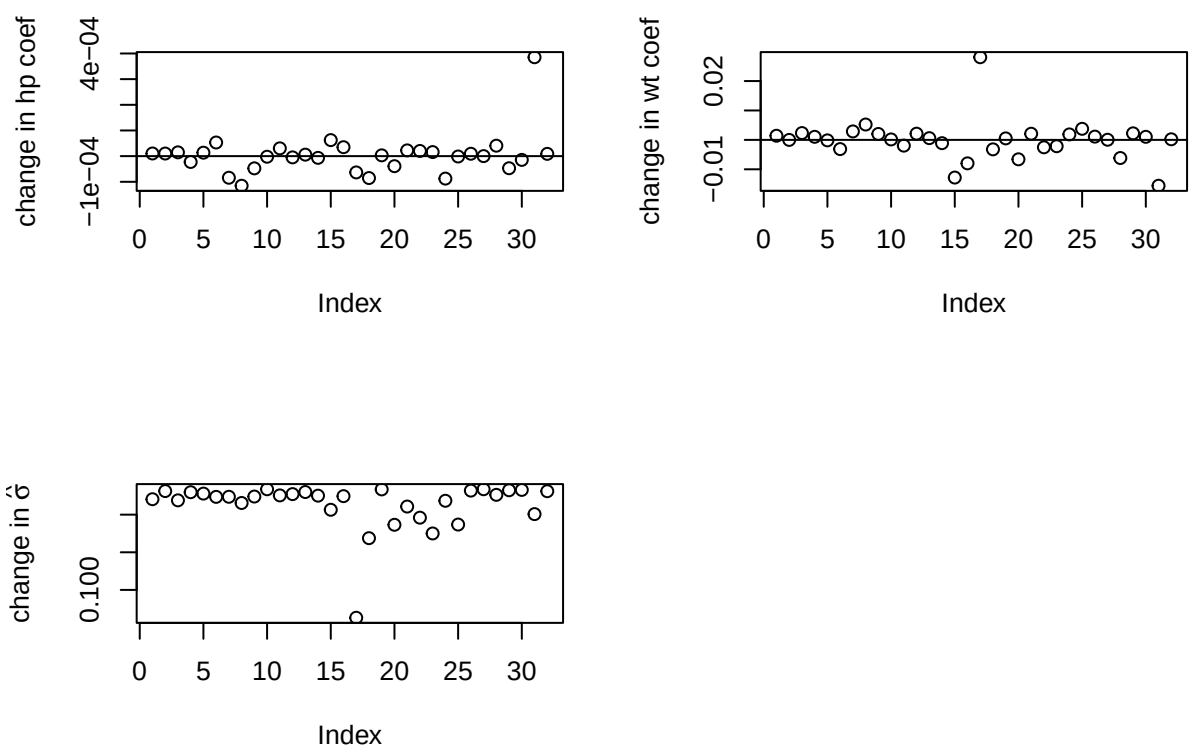

Figure 15: Changes on coefficient estimates due to each observation

Figure 15 shows how each coefficient and $\hat{\sigma}$ would change when leaving one of the observations out to fit the model. The larger the change means that the observation that is left out would have higher influence on the parameter estimate. We see that there indeed seems to exist some influential points for the parameter estimates. Table 11 summarizes which observation might have relatively high influence on the parameter estimates, where "*" denotes that the corresponding observation is among the first three most influential points for the corresponding parameter estimate.

Table 10: Summary for the influential points

	Maserati.Bora	Cadillac.Fleetwood	Valiant	Chrysler.Imperial
hp				
wt				*
$\hat{\sigma}$				

Table 11: Summary for the influential points (Cont.)

	Merc.240D	Pontiac.Firebird	Porsche.914-2	Merc.280	Honda.Civic
hp					
wt	*	*			
$\hat{\sigma}$			*	*	*

```
summ = matrix("", ncol = length(unique(unlist(inf.ls))),
              nrow = length(variables) + 1)
colnames(summ) = unique(unlist(inf.ls))
rownames(summ) = c(variables, "$\\hat{\\sigma}$")
for(i in 1:length(variables) + 1){
  fill = rep("", ncol(summ))
  fill[unique(unlist(inf.ls)) %in% inf.ls[[i]]] = "*"
  summ[i,] = fill
}
kable(summ[,1:4], caption = "Summary for the influential points",
      escape = FALSE, booktabs = TRUE, align = "c")
```

```
kable(summ[,5:9], caption = "Summary for the influential points (Cont.)",
      escape = FALSE, booktabs = TRUE, align = "c")
```

Though there seems to be some sample points having relatively large influences, we choose not to remove any of them, considering that the change on the coefficient estimates due to their influences does not seem really big compared to the coefficient estimates obtained with the full data.

Diagnostic for nonconstant variance

```
lev = lm.influence(fit2)$hat
rstud = fit2$res/(summary(fit2)$sigma*(1-lev))
par(mfrow = c(2, 2))
plot(fit2$res,ylab="raw residuals")
plot(rstud,ylab="studentized residuals")
plot(jack,ylab="Jackkniife residuals")
```

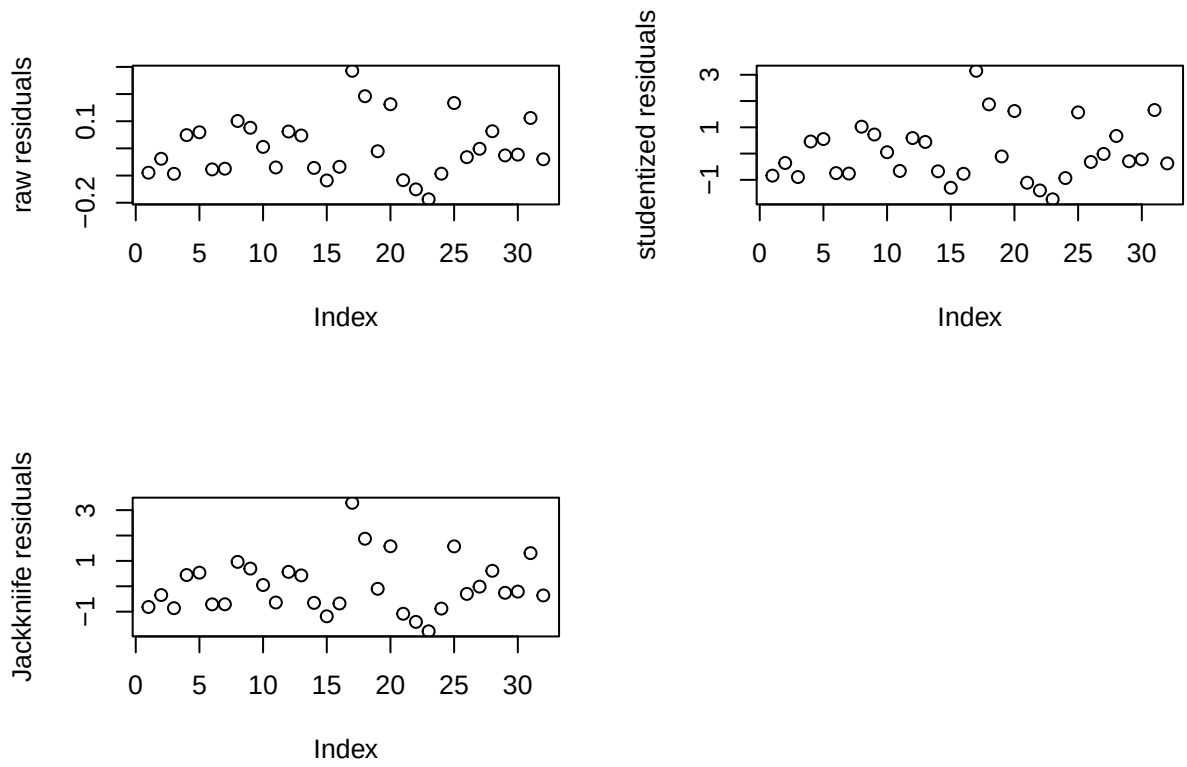

Figure 16: Residual plots using raw residuals, studentized residuals and Jackknife residuals

Figure 16 shows that the patterns of the three kinds of residual plots look similar. Therefore, we decide to use the raw residuals in the following analysis. To increase the resolution of detecting non-constant variance, the plot of absolute residuals against fitted values is shown in Figure 17.

```
plot(abs(fit2$res)~fit2$fitted.values, ylab="|raw residuals|", xlab = "fitted value")
```

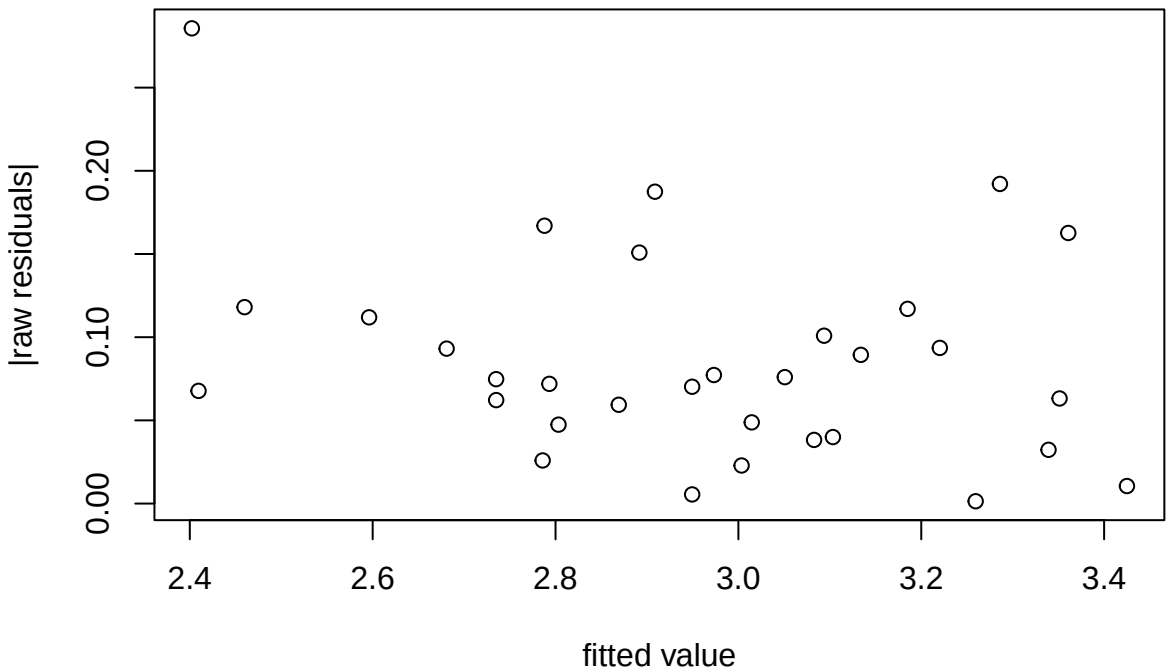

Figure 17: Absolute residuals against fitted values

It seems that there needs no worry about the problem of nonconstant variance. We can roughly make a quick check for it by regressing the absolute residuals on the fitted values. The fitting result for the regression just described is shown in Table 12. We can see that the coefficient of "fitted values" is not significant under 0.05 significance level. Hence, we conclude that the assumption for constant variance might be acceptable.

```
nonconst.fit = lm(abs(fit2$res)~fit2$fitted.values)
summ.table(nonconst.fit, cap = "Test for nonconstant variance by regression",
           rows = c("(Intercept)", "fitted values"))
```

Table 12: Test for nonconstant variance by regression

	Estimate	SE	t-value	p-value
(Intercept)	0.2762	0.1170	2.3609	0.0249
fitted values	-0.0642	0.0394	-1.6291	0.1137

Diagnostic for curvature in the mean of residuals

```
par(mfrow = c(1, 2))

d = lm(log(mpg)~wt, data = dat)$res
m = lm(hp~wt, data = dat)$res
```

```
plot(m, d, xlab = "hp residuals", ylab = "mpg residual", main = "added residual plot")

d = lm(log(mpg)~hp, data = dat)$res
m = lm(wt~hp, data = dat)$res
plot(m, d, xlab = "wt residuals", ylab = "mpg residual", main = "added residual plot")
```

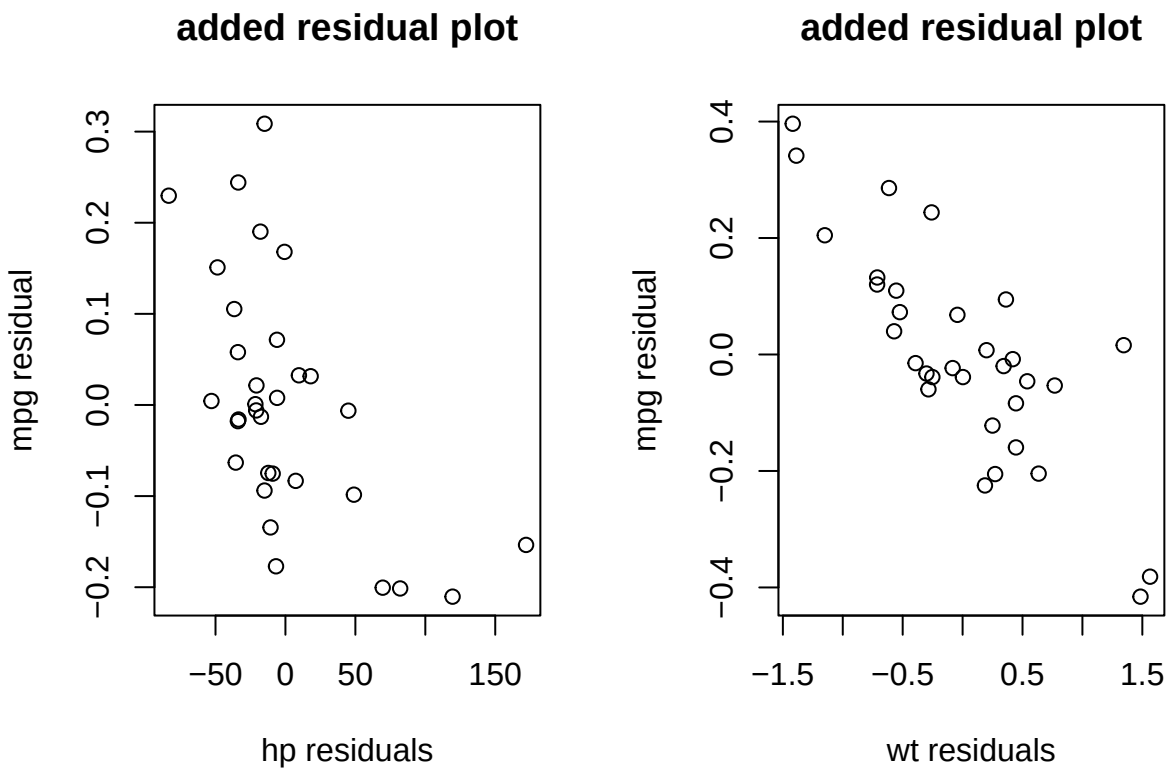

Figure 18: Added variable plots

Figure 18 shows the added residual plot for each predictor. Since each plot shows that there seems to exist only linear relationship between the two kinds of residuals, we conclude that there might not be curvature in the mean of residuals.

Diagnostic for Normality

```
qqnorm(fit2$res, ylab = "raw residuals", main = "")
qqline(fit2$res)
```

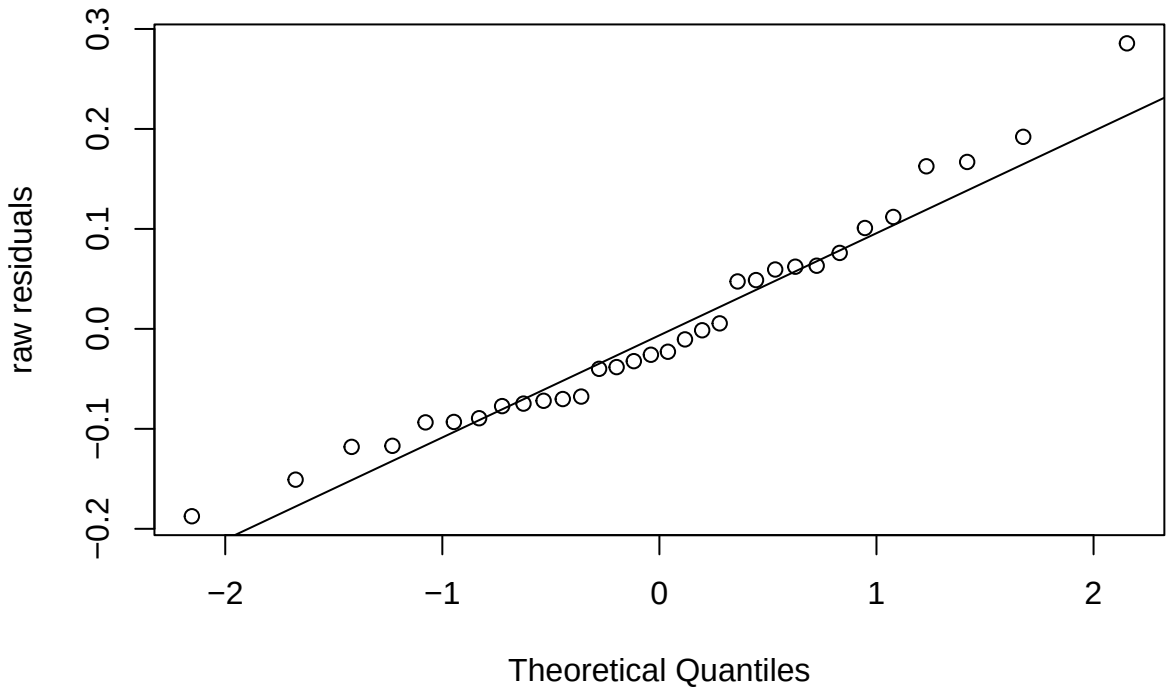

Figure 19: Normal Q-Q plot

Q-Q plot is used to check for normality. In Figure 19, since all the points lie quite close to the line joining the first and the third quartiles, we conclude that there is no strong evidence to say that the normality assumption does not hold. In other words, normality might be an acceptable assumption.

1.4.3 Summary for the diagnostics

- Though there are some leverage points, outliers or influential points for the two models, they all do not harm so much in terms of the stability of the model fittings.
- Note that constant variance, linearity and normality are three crucial assumption in linear model. Model (1) conflicts with the constant variance assumption, while Model (2) meets all the assumptions. Therefore inference based on Model (2) is supposed to be more valid than that based on Model (1).

1.5 Some predictions of future observations for interesting values of the predictors

1.5.1 Prediction by Model (1)

For me, it is interesting to know how well the endurance would be in terms of `mpg` for the automobile with the lowest weight (1.51 1000 lbs), the highest quarter mile time (22.9) and manual transmission simultaneously. Note the so-called lowest weight and highest quarter mile time are limited to those observed in this data, while

we have no such observation with the lowest weight, the highest quarter mile time and manual transmission at the same time.

```
pred = predict(fit1, newdata = data.frame(wt = 1.51, qsec = 22.9, am = factor(1)),
               interval = "prediction")
```

It turns out that the predicted value is 34.7124858 and the 95% predictive interval is [28.7388365, 40.6861351].

1.5.2 Prediction by Model (2)

For me, it is also interesting to know how well the endurance would be in terms of `mpg` for the automobile with the highest horsepower (335) and the lowest weight (1.51 1000 lbs) simultaneously. Note the so-called highest horsepower and lowest weight are limited to those observed in this data, while we have no such observation with the highest horsepower and the lowest weight at the same time.

```
pred = predict(fit2, newdata = data.frame(hp = 335, wt = 1.51), interval = "prediction")
```

It turns out that the predicted value is 3.0092212 and the 95% predictive interval is [2.6874332, 3.3310093].

1.6 An interpretation of the meaning of the model (parameters) with respect to the particular area of application

1.6.1 Interpretation for Model (1)

Recall Model (1) and Table 2, we have $\hat{\beta}_0' = 9.6177805$, $\hat{\beta}_1' =$ -3.9165037, $\hat{\beta}_2' = 1.225886$ and $\hat{\beta}_3' = 1.225886$. Such result can be interpreted as follows.

- Each 1000 lbs increase in weight is associated with 3.9165037 miles per gallon decrease in endurance.
- Each unit increase in quarter mile time is associated with 1.225886 miles per gallon increase in endurance.
- Automobiles with manual transmission is 2.9358372 miles per gallon higher in endurance than those with automatic transmission.

1.6.2 Interpretation for Model (2)

Recall Model (2) and Table 4, we have $\hat{\gamma}_0' = 3.829103$, $\hat{\gamma}_1' =$ -0.0015435 and $\hat{\gamma}_2' = 1.225886$. Such result can be interpreted as follows.

- Each unit increase in gross horsepower is associated with 0.0015435 miles per gallon decrease in endurance.
- Each 1000 lbs increase in weight is associated with 0.2005368 miles per gallon decrease in endurance.

2 Problem 2: Identify each variable as nominal, ordinal, or interval

a. UK political party preference (Labour, Liberal Democrat, Conservative)

I would identify this variable as nominal when the political ideology lying behind the preference is ignorable or is not utilized in the analysis. If the information regarding ideology is meaningful and essential for the analysis, I would identify it as ordinal according to the some aspects of ideology such as the conservativeness.

b. Anxiety rating (none, mild, moderate, severe, very severe)

Since it is obvious that the levels of rating is ordinal and the information provided is usually quite essential and meaningful, I would identify this variable as ordinal.

c. Patient survival (in number of months)

We know that the time of patient survival must be a continuous variable, while the variable described here is recorded by month. This implies that the information regarding survival time that can be provided by this variable is at most an interval containing the patient's dying day. Taking for example a patient recorded died in the 8th month, the information we can get is nothing more than the fact that he/she died on some day during the 8th month. Hence, I would identify this variable as interval.

d. Clinic location (London, Boston, Madison, Rochester, Montreal)

There doesn't seem to be possibility that information more than the nominal aspect can be provided by this variable. Therefore, I would identify it as nominal.

e. Response of tumor to chemotherapy (complete elimination, partial reduction, stable, growth progression)

It is clear that there is information regarding the efficacy of the chemotherapy. Therefore, I would identify it as ordinal.

f. Favorite grocery store for UK residents (Sainsbury, Tesco, Waitrose, other)

It seems that no clear information other than the nominal aspect can be provided by this variable. Therefore, I would identify it as nominal.

3 Problem 3: Each of 100 multiple-choice questions on an exam has four possible answers, one of which is correct. For each question, a student guesses by selecting an answer randomly.

a. Specify the distribution of the number of correct answer.

Due to the random selection, the probability of guessing correctly for each question is $\frac{1}{4}$. Moreover, the guessing process is supposed to be mutually independent for each question. Therefore, we can regard the guessing result for each question as a bernoulli trial, and thus the distribution of the number of correct answer is clearly a binomial distribution with number of trials $n = 100$ and success probabiltiy $p = \frac{1}{4}$.

b. Find the mean and standard deviation of that distribution. Would it be surprising if the student made at least 50 correct responses? Why?

Denote by X the number of correct responses. As illustrated in a., we have $X \sim Bin(n = 100, p = \frac{1}{4})$ By definition, the pmf of X is

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Then the mgf of X can be derived as follows.

$$\begin{aligned} M(t) &= E(e^{tX}) \\ &= \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x (1-p)^{n-x} \\ &= (1-p+pe^t)^n. \end{aligned}$$

Note that $M(t)$ exists for an interval centered at $t = 0$.

- The mean of the distribution is $np = 100 \times \frac{1}{4} = 25$.

Proof.

$$\begin{aligned} E(X) &= M'(0) \\ &= n(1-p+pe^t)^{n-1} pe^t \mid_{t=0} \\ &= np. \end{aligned}$$

□

- The standard deviation of the distribution is $\sqrt{np(1-p)} = \sqrt{100 \times \frac{1}{4} \times \left(1 - \frac{1}{4}\right)} = 4.330127$.

Proof.

$$\begin{aligned} E(X^2) &= M''(0) \\ &= n\{(n-1)(1-p+pe^t)^{n-2}(pe^t)^2 + (1-p+pe^t)^{n-1} pe^t\} \mid_{t=0} \\ &= n(n-1)p^2 + np \end{aligned}$$

$$\begin{aligned} var(X^2) &= E(X^2) - E(X)^2 \\ &= n(n-1)p^2 + np - (np)^2 \\ &= np(1-p) \implies SD(X) = \sqrt{np(1-p)} \end{aligned}$$

□

- Using R, it is easy to obtain that $P(X \geq 50) \approx 6.64 \times 10^{-8}$, which is extremely small. Therefore, I would say that it would be very surprising if the student made at least 50 correct responses.

```
prob.at.least.50 = 1-pbinom(49, size = 100, prob = .25)
prob.at.least.50
```

```
## [1] 6.638502e-08
```

c. Specify the distribution of (n_1, n_2, n_3, n_4), wheren_j is the number of times the student picked choice j.

Denote by p_j the probability of picking choice j. It is trivial to specify the distribution of (n_1, n_2, n_3, n_4) by a multinomial distribution with $n = 100$, and $(n_1, n_2, n_3, n_4) = (0.25, 0.25, 0.25, 0.25)$. Note that $\sum_{j=1}^{4} n_j = 100$ and $\sum_{j=1}^{4} p_j = 1$. The pdf for the distribution is

$$f(n_1, n_2, n_3, n_4) = \frac{n!}{n_1!n_2!n_3!n_4!} p_1^{n_1} p_2^{n_2} p_3^{n_3} p_4^{n_4},$$

with n_1, n_2, n_3, n_4 all non-negative integer-valued.

d. Find $E(n_j)$ and $var(n_j)$. Show that $cov(n_j, n_k) = -6.25$ and $cor(n_j, n_k) = -0.333$.

- It is trivial that n_j follows binomial distribution with $n = 100$ and $p = 0.25$ (see the proof below). By b., it is readily obtained that $E(n_j) = 25$ and $var(n_j) = 18.75$.

Proof. WLOG, prove only the result for n_1. Denote by $h(n_1)$ the pdf of n_1.

$$\begin{aligned}
h(n_1) &= \sum_{n_2,n_3,n_4} f(n_1, n_2, n_3, n_4) \\
&= \sum_{n_2,n_3,n_4} \frac{n!}{n_1!n_2!n_3!n_4!} p_1^{n_1} p_2^{n_2} p_3^{n_3} p_4^{n_4} \\
&= \frac{n!}{n_1!(n-n_1)!} p_1^{n_1} (1-p_1)^{n-n_1} \sum_{n_2,n_3,n_4} \frac{(n-n_1)!}{n_2!n_3!n_4!} \left(\frac{p_2}{1-p_1}\right)^{n_2} \left(\frac{p_3}{1-p_1}\right)^{n_3} \left(\frac{p_4}{1-p_1}\right)^{n_4} \\
&= \binom{n}{n_1} p_1^{n_1} (1-p_1)^{n-n_1},
\end{aligned}$$

which is the form of the pmf of binomial distribution. □

- By the fact that $n_j \mid n_k \sim Bin\left(n = 100 - n_k, p = \frac{1}{3}\right)$. We can show that

$$\begin{aligned}
cov(n_j, n_k) &= E(n_j n_k) - E(n_j)E(n_k) \\
&= E\{n_k E(n_j|n_k)\} - E(n_j)E(n_k) \\
&= E\left\{\frac{1}{3}(100 - n_k)n_k\right\} - E(n_j)E(n_k) \\
&= \frac{100E(n_k)}{3} - \frac{E(n_k)^2 + var(n_k)}{3} - E(n_j)E(n_k) \\
&= \frac{100 \times 25}{3} - \frac{25^2 + 18.75}{3} - 25^2 \\
&= -6.25.
\end{aligned}$$

Moreover, we have

$$cor(n_j, n_k) = \frac{cov(n_j, n_k)}{\sqrt{var(n_j)} \times \sqrt{var(n_k)}} = \frac{-6.25}{\sqrt{18.75} \times \sqrt{18.75}} \approx -0.333.$$

4 Appendix

4.1 A

```
fit = lm(mpg~.-cyl, dat)
summ.table(fit)
```

Table 13: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	10.9601	13.5303	0.8100	0.4266
disp	0.0128	0.0168	0.7626	0.4538
hp	-0.0219	0.0209	-1.0477	0.3062
drat	0.8352	1.5363	0.5437	0.5921
wt	-3.6925	1.8395	-2.0073	0.0572
qsec	0.8424	0.6868	1.2267	0.2329
vs1	0.3897	1.9480	0.2001	0.8433
am1	2.5774	1.9403	1.3283	0.1977
gear	0.7116	1.3656	0.5210	0.6075
carb	-0.2196	0.7886	-0.2785	0.7833

```
fit = update(fit, .~.-vs); summ.table(fit)
```

Table 14: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	9.7683	11.8923	0.8214	0.4199
disp	0.0121	0.0161	0.7532	0.4590
hp	-0.0210	0.0199	-1.0514	0.3040
drat	0.8751	1.4911	0.5869	0.5630
wt	-3.7115	1.7983	-2.0639	0.0505
qsec	0.9108	0.5831	1.5620	0.1319
am1	2.5239	1.8813	1.3416	0.1928
gear	0.7598	1.3158	0.5775	0.5692
carb	-0.2480	0.7593	-0.3266	0.7470

```
fit = update(fit, .~.-carb); summ.table(fit)
```

Table 15: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	9.1976	11.5422	0.7969	0.4333
disp	0.0155	0.0121	1.2782	0.2134
hp	-0.0247	0.0160	-1.5478	0.1348
drat	0.8102	1.4501	0.5588	0.5815
wt	-4.1307	1.2359	-3.3421	0.0027
qsec	1.0098	0.4888	2.0657	0.0498
am1	2.5898	1.8353	1.4111	0.1710
gear	0.6064	1.2060	0.5029	0.6196

```
fit = update(fit, .~.-gear); summ.table(fit)
```

Table 16: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	10.7106	10.9754	0.9759	0.3385
disp	0.0131	0.0110	1.1930	0.2441
hp	-0.0218	0.0147	-1.4875	0.1494
drat	1.0207	1.3675	0.7464	0.4624
wt	-4.0445	1.2056	-3.3548	0.0025
qsec	0.9907	0.4800	2.0639	0.0496
am1	2.9847	1.6338	1.8268	0.0797

```
fit = update(fit, .~.-drat); summ.table(fit)
```

Table 17: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	14.3619	9.7408	1.4744	0.1524
disp	0.0112	0.0106	1.0598	0.2990
hp	-0.0212	0.0145	-1.4596	0.1564
wt	-4.0843	1.1941	-3.4204	0.0021
qsec	1.0069	0.4754	2.1179	0.0439
am1	3.4705	1.4858	2.3358	0.0275

```
fit = update(fit, .~.-disp); summ.table(fit)
```

Table 18: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	17.4402	9.3189	1.8715	0.0721
hp	-0.0176	0.0142	-1.2471	0.2231
wt	-3.2381	0.8899	-3.6387	0.0011
qsec	0.8106	0.4389	1.8470	0.0757
am1	2.9255	1.3971	2.0939	0.0458

```
fit1 = update(fit, .~.-hp); summ.table(fit)
```

Table 19: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	17.4402	9.3189	1.8715	0.0721
hp	-0.0176	0.0142	-1.2471	0.2231
wt	-3.2381	0.8899	-3.6387	0.0011
qsec	0.8106	0.4389	1.8470	0.0757
am1	2.9255	1.3971	2.0939	0.0458

4.2 B

```
fit = lm(log(mpg)~.-cyl, dat)
summ.table(fit)
```

Table 20: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	2.8679	0.6141	4.6701	0.0001
disp	0.0001	0.0008	0.1109	0.9127
hp	-0.0009	0.0009	-0.9136	0.3708
drat	0.0189	0.0697	0.2710	0.7889
wt	-0.1739	0.0835	-2.0827	0.0491
qsec	0.0293	0.0312	0.9399	0.3575
vs1	-0.0078	0.0884	-0.0884	0.9303
am1	0.0434	0.0881	0.4934	0.6267
gear	0.0554	0.0620	0.8937	0.3811
carb	-0.0187	0.0358	-0.5234	0.6059

```
fit = update(fit, .~.-vs); summ.table(fit)
```

Table 21: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	2.8918	0.5394	5.3616	0.0000
disp	0.0001	0.0007	0.1346	0.8941
hp	-0.0009	0.0009	-0.9808	0.3369
drat	0.0181	0.0676	0.2676	0.7914
wt	-0.1735	0.0816	-2.1273	0.0443
qsec	0.0279	0.0264	1.0559	0.3020
am1	0.0445	0.0853	0.5218	0.6068
gear	0.0544	0.0597	0.9120	0.3712
carb	-0.0182	0.0344	-0.5274	0.6030

```
fit = update(fit, .~.-disp); summ.table(fit)
```

Table 22: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	2.9220	0.4804	6.0825	0.0000
hp	-0.0008	0.0007	-1.1370	0.2668
drat	0.0183	0.0662	0.2761	0.7849
wt	-0.1647	0.0479	-3.4381	0.0021
qsec	0.0259	0.0212	1.2209	0.2340
am1	0.0436	0.0833	0.5237	0.6053
gear	0.0540	0.0584	0.9251	0.3641
carb	-0.0211	0.0259	-0.8165	0.4223

```
fit = update(fit, .~.-drat); summ.table(fit)
```

Table 23: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	2.9685	0.4415	6.7239	0.0000
hp	-0.0008	0.0007	-1.2070	0.2387
wt	-0.1677	0.0458	-3.6625	0.0012
qsec	0.0267	0.0206	1.2954	0.2070
am1	0.0490	0.0795	0.6155	0.5438
gear	0.0575	0.0559	1.0292	0.3132
carb	-0.0200	0.0251	-0.7977	0.4325

```
fit = update(fit, .~.-am); summ.table(fit)
```

Table 24: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	3.0332	0.4237	7.1594	0.0000
hp	-0.0009	0.0007	-1.3167	0.1994
wt	-0.1754	0.0435	-4.0309	0.0004
qsec	0.0225	0.0192	1.1703	0.2525
gear	0.0754	0.0472	1.5969	0.1224
carb	-0.0206	0.0248	-0.8332	0.4123

```
fit = update(fit, .~.-carb); summ.table(fit)
```

```
fit = update(fit, .~.-gear); summ.table(fit)
```

```
fit2 = update(fit, .~.-qsec); summ.table(fit)
```

Table 25: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	3.0824	0.4171	7.3893	0.0000
hp	-0.0011	0.0006	-1.6779	0.1049
wt	-0.1916	0.0388	-4.9424	0.0000
qsec	0.0260	0.0186	1.3999	0.1729
gear	0.0502	0.0361	1.3919	0.1753

Table 26: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	3.3881	0.3605	9.3989	0.0000
hp	-0.0009	0.0006	-1.4092	0.1698
wt	-0.2226	0.0322	-6.9073	0.0000
qsec	0.0234	0.0188	1.2456	0.2232

Table 27: Model Summary

	Estimate	SE	t-value	p-value
(Intercept)	3.3881	0.3605	9.3989	0.0000
hp	-0.0009	0.0006	-1.4092	0.1698
wt	-0.2226	0.0322	-6.9073	0.0000
qsec	0.0234	0.0188	1.2456	0.2232